Project name: Talk to Character

Tracks: Conversation as the Interface (primary); Put a Face to the Voice (secondary); Agents That Act (supporting tool use).

One-sentence description: Pause a movie and talk face-to-face with its characters in their own voices, choosing whether they know the ending or only what has happened so far.

Project write-up

Talk to Character lets movie viewers pause a film and talk to the characters on screen. It's for people who want to explore a character's perspective without searching through plot summaries and risking spoilers. Most character chatbots have access to the whole story, so their answers can reveal events the viewer hasn't reached. Our conversations follow the pause point, using subtitles and screenplay information to build memories and track each character's relationships and emotional state. In Present mode, replies draw on what the character has experienced so far, with an additional check to help catch spoilers. Viewers can switch to Omniscient mode at any time to discuss the full story, including the ending. To start, they pause a prepared film, select a character, and speak or type in their preferred language. Higgs Realtime generates spoken replies in a cloned character voice and supports interruptions. An Higgs Avatar mode animates the character's face using the reply audio. As viewers move through the film, the character's available memories and perspective change with it.

Team Information

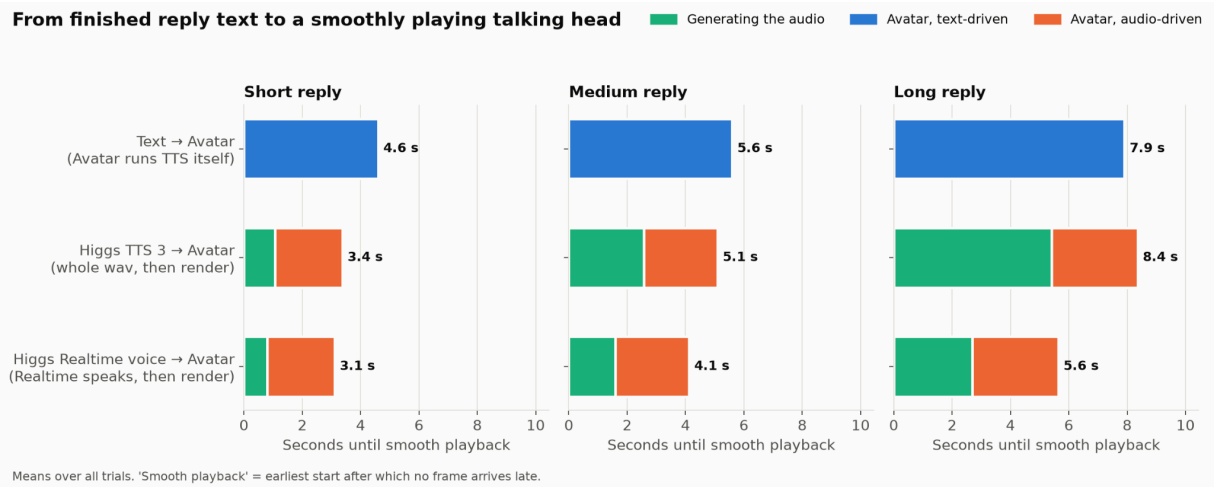
Members:

- Haochen Ding (@Dennis-Ding1): Backend development, Higgs API integration, and character knowledge processing.
- Rolland He (@Rolland-He): Real-time voice interaction, avatar integration, and testing.
- Yinuo Yang (@CiaraYang): Frontend interface and user experience, including the movie player, character selection, and conversation panel.

Repository: https://github.com/Dennis-Ding1/Talk_to_Character.git

Known limitations:

- Avatar mode adds latency because speech must be generated, checked for spoilers, and rendered into video before playback. In our tests, Realtime audio-to-avatar took roughly 3–6 seconds to reach smooth playback, while sending text directly to Avatar took around 5–8 seconds. Longer replies generally take more time.



- Face detection and avatar generation are less reliable for anime characters and heavily stylized faces. Clearly visible eyes, a nose, and a mouth are needed; exaggerated proportions or artistic effects can cause detection failures or poor animation.